LESSONS LEARNED

Student Performance Analytics Dashboard

Project Name: Student Performance Analytics Dashboard
Organization: Mega Central High School (Case Study)
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Project Overview:

The Student Performance Analytics Dashboard was developed to provide school stakeholders with actionable insights into student performance, identify academically at-risk students, and support data-driven decision-making.

Throughout the project lifecycle, several valuable lessons were learned regarding business analysis, data modeling, dashboard design, stakeholder requirements, and project execution.

Key Lessons Learned:

1. Business Requirements Must Be Clearly Defined:

Initially, the project scope focused only on student marks. During requirement analysis, it became evident that stakeholders required much more than score reporting, including:

- Student risk identification
- Subject-level analysis
- Chapter-level insights
- Executive KPIs
- Individual student performance tracking

Lesson Learned

Investing sufficient time in requirement elicitation reduces rework during development.

2. Data Quality Determines Dashboard Quality:

Several issues were encountered while generating synthetic datasets, including:

- Duplicate records
- Missing relationships
- Inconsistent marks
- Unrealistic test structures

These issues impacted report accuracy until the datasets were cleaned and standardized.

Lesson Learned

High-quality dashboards depend on clean, well-structured data.

3. Data Modeling is Critical:

Creating proper relationships between:

- Students
- Tests
- Responses
- Question Bank
- Scores

was one of the most important aspects of the project.
Improper relationships initially produced incorrect calculations and filter behavior.
Lesson Learned
A well-designed data model simplifies report development and improves accuracy.

4. Keep Dashboards Simple:
Early dashboard designs contained too many visuals on a single page, making interpretation difficult.
The final design focused on:

- Clear KPIs
- Minimal colors
- Consistent layout
- Logical grouping of visuals

Lesson Learned
Simple dashboards improve usability and decision-making.

5. Stakeholder Perspective Matters:
Different stakeholders required different information.
For example:
Principal

- School performance
- Risk overview

Teachers

- Student details
- Weak chapters

Academic Coordinator

- Subject analytics
- Curriculum insights

Lesson Learned
Dashboards should be designed around user needs rather than available data.

6. Business Analysis is More Than Documentation:
This project reinforced that Business Analysis involves:

- Understanding business problems
- Asking the right questions

- Defining requirements
- Validating solutions
- Supporting testing
- Facilitating decision-making

Technical Lessons:

- Power BI relationships should follow a star schema wherever possible.
- Measures provide greater flexibility than calculated columns.
- Consistent naming conventions improve maintainability.
- Date tables are essential for time-based analysis.
- KPI cards should highlight actionable information rather than excessive metrics.

Personal Learnings:

Through this project, I gained practical experience in:

- Requirement Gathering
- Stakeholder Analysis
- Data Modeling
- Dashboard Design
- User Acceptance Testing
- Business Documentation
- Power BI Development
- SQL-based Data Preparation

Conclusion:

This project strengthened my understanding of the end-to-end Business Analysis lifecycle and demonstrated how analytical dashboards can support educational institutions in improving student outcomes.